Combinatorics (Additive, extremal, graph, yayayayayay)

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Yeah so a lot of combinatorics is about finding valid constructions. Actually this is a trend in all of math, so a background in what a construction is is great.

1 Set Theory

2 Extremal Combinatorics

Definition

Sunflower A sunflower is a collection of sets $S_1, S_2, \ldots, S_k$ such that $S_i \cap S_j$ is constant over all i, j.

It's called a sunflower because it looks like one when you draw it:

Theorem

Erdos-Rado Sunflower Lemma Given a family $\mathcal{F}$ of size s sets, we may find a sunflower of size k if $|\mathcal{F}| > s!(k-1)k^{s-1}$

Proof

Define $f(s,k)$ to be the maximal cardinality of $\mathcal{F}$ such that it does not contain a sunflower. Continue with induction on s, and suppose that no sunflower exists in $\mathcal{F}$. If $s = 1$, clearly $f(1,k) \leq k-1$. To induct $s-1 \to s$, let $A_1, \ldots, A_t$ be the maximum collection of pairwise disjoint sets in $\mathcal{F}$. Then define $A = \cup_{1 \leq i \leq t} A_i$ so that $|A| < sk$, or else we can select k disjoint sets. Define $\mathcal{F}_a = \{S \not\ni a : a \in S, S \in \mathcal{F}\}$. Bound $|\mathcal{F}| \leq \sum_a |\mathcal{F}_a| \leq f(s-1,k) \cdot sk \leq s!(k-1)k^{s-1}$ as desired.

Problem

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